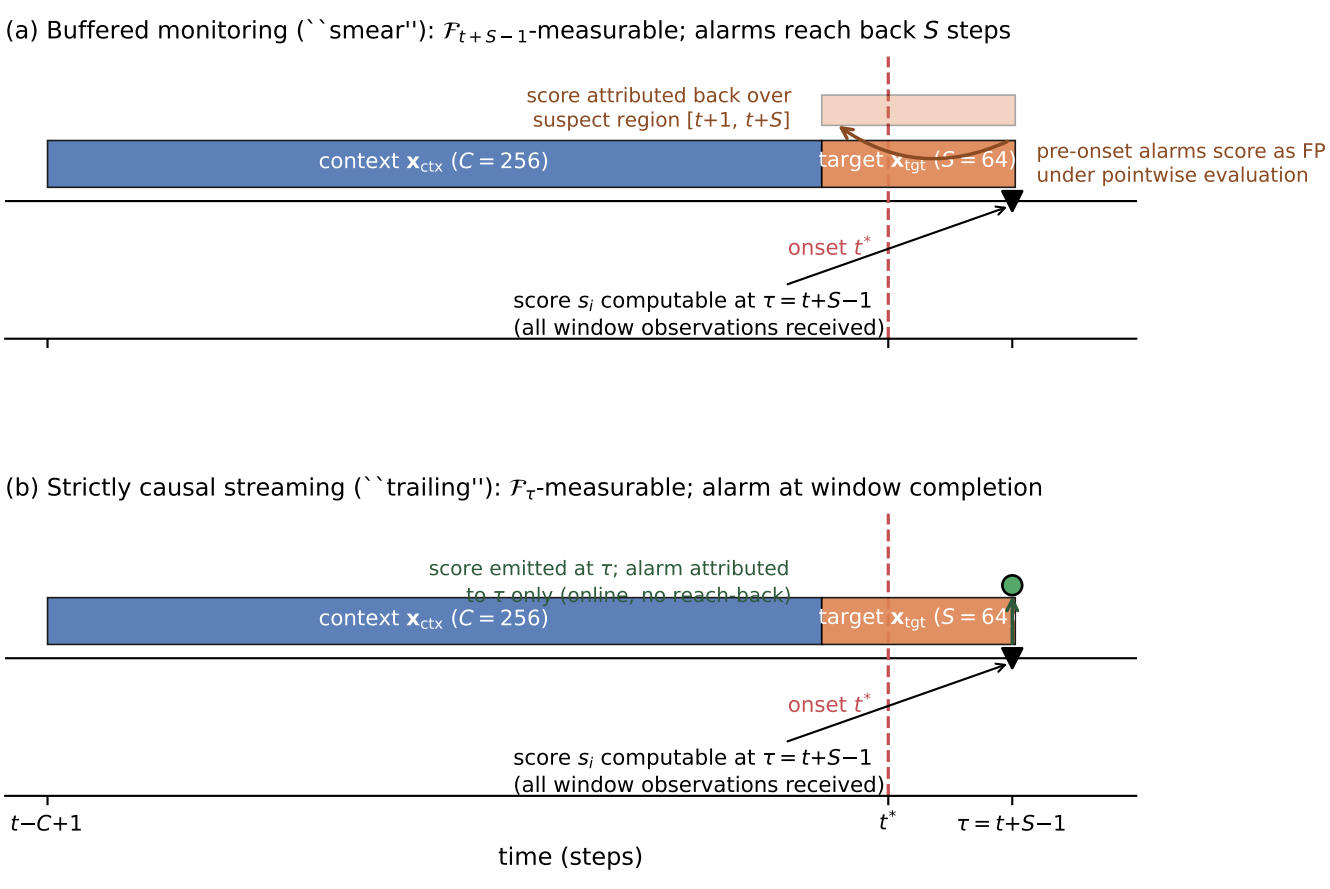
(a) Buffered monitoring (``smear''): $\mathcal{F}_{t+S-1}$-measurable; alarms reach back S steps
score attributed back over suspect region [t+1, t+S]
context $\mathbf{x}_{\text{ctx}}$ ($C = 256$)
target $\mathbf{x}_{\text{tgt}}$ ($S = 64$)
pre-onset alarms score as FP under pointwise evaluation
onset t^*
score s_i computable at $\tau = t+S-1$ (all window observations received)
(b) Strictly causal streaming (``trailing''): $\mathcal{F}_\tau$-measurable; alarm at window completion
score emitted at τ; alarm attributed to τ only (online, no reach-back)
context $\mathbf{x}_{\text{ctx}}$ ($C = 256$)
target $\mathbf{x}_{\text{tgt}}$ ($S = 64$)
onset t^*
score s_i computable at $\tau = t+S-1$ (all window observations received)
$t-C+1$
t^*
$\tau = t+S-1$
time (steps)